FEDERAL BOARD OF INTERMEDIATE & SECONDARY EDUCATION

Revision Test 2026

Physics — Class 9 (Science Group)

Syllabus: Unit 1 — Physical Quantities & Measurement | Unit 2 — Kinematics | Unit 3 — Dynamics-I | Unit 4 — Dynamics-II | Unit 5 — Pressure & Deformation in Solids | Unit 6 — Work & Energy (till Ex. 6.4 only) | Unit 9 — Nature of Science & Physics

Total Marks: 43 **Time: 2 Hours** **Paper: Exam-14 (Version 1)**

1. Attempt ALL questions. There are no choices in this paper.
2. Section A (24 marks): 24 MCQs. Fill the correct bubble. Suggested time: 30 minutes.
3. Show complete working for every question that requires it. Always write the correct unit with your final answer.
4. Use black or blue ink only. Pencil is allowed only for diagrams. Take g = 10 m/s^2 unless stated otherwise.

SECTION A — Multiple Choice Questions (24 × 1 = 24 Marks)

#	Q.1 Choose the best possible option.	A	B	C	D	Answer
1	The least count of a screw gauge is found by dividing its pitch by the:	number of main scale divisions	total length of the scale	number of circular scale divisions	smallest main scale division	ⒶⒷⒸⒹ
2	Written in scientific notation, 0.00085 m is:	8.5×10^{-4} m	8.5×10^{-3} m	85×10^{-3} m	0.85×10^{-3} m	ⒶⒷⒸⒹ
3	Which one of the following is a vector quantity?	mass	temperature	energy	momentum	ⒶⒷⒸⒹ
4	The SI prefix that represents 10^{-9} is:	micro	milli	nano	mega	ⒶⒷⒸⒹ
5	The area under a speed-time graph gives the:	acceleration	distance travelled	average speed	force applied	ⒶⒷⒸⒹ
6	For a body falling freely near the surface of the Earth, the acceleration is approximately:	9.8 m/s^2	1.6 m/s^2	98 m/s^2	zero	ⒶⒷⒸⒹ
7	When the velocity of a body changes by unequal amounts in equal intervals of time, its acceleration is said to be:	uniform	zero	constant	non-uniform	ⒶⒷⒸⒹ
8	A car covers a distance of 96 m in 6 s. Its average speed is:	12 m/s	16 m/s	24 m/s	576 m/s	ⒶⒷⒸⒹ
9	According to Newton's second law of motion, the acceleration produced in a body is:	directly proportional to its mass	inversely proportional to the applied force	directly proportional to the applied force	independent of the applied force	ⒶⒷⒸⒹ
10	The SI unit of impulse is:	newton (N)	watt (W)	joule (J)	newton second (N s)	ⒶⒷⒸⒹ
11	Newton's laws of motion cannot be readily applied to objects on the:	atomic scale	scale of a moving car	scale of a falling stone	scale of a rolling ball	ⒶⒷⒸⒹ
12	The laboratory instrument used to measure the mass of a body is the:	force meter	electronic balance	measuring cylinder	stopwatch	ⒶⒷⒸⒹ
13	The single point at which the whole weight of a body appears to act is called its:	centre of curvature	axis of rotation	centre of gravity	line of action	ⒶⒷⒸⒹ
14	The average orbital speed of a satellite moving in a circular orbit of radius r with orbital period T is given by:	$v = 2\pi r / T$	$v = T / 2\pi r$	$v = 2\pi rT$	$v = \pi r^2 / T$	ⒶⒷⒸⒹ

15	Two parallel forces acting in the same direction are called:	unlike parallel forces	action-reaction forces	balanced forces	like parallel forces	Ⓐ Ⓑ Ⓒ Ⓓ
16	According to the principle of moments, a body is in equilibrium when the sum of the clockwise moments is:	greater than the sum of the anticlockwise moments	equal to the sum of the anticlockwise moments	less than the sum of the anticlockwise moments	always zero	Ⓐ Ⓑ Ⓒ Ⓓ
17	The SI unit of the spring constant is:	newton per metre (N/m)	newton (N)	newton metre (N m)	metre per newton (m/N)	Ⓐ Ⓑ Ⓒ Ⓓ
18	On a load-extension graph, the point beyond which the extension is no longer directly proportional to the load is called the:	breaking point	zero error	limit of proportionality	least count	Ⓐ Ⓑ Ⓒ Ⓓ
19	The instrument used to measure atmospheric pressure is the:	manometer	barometer	force meter	screw gauge	Ⓐ Ⓑ Ⓒ Ⓓ
20	At sea level, atmospheric pressure supports a mercury column of height about:	13.6 cm	76 mm	7.6 cm	76 cm	Ⓐ Ⓑ Ⓒ Ⓓ
21	Which one of the following is a non-renewable energy resource?	wind	coal	tidal	geothermal	Ⓐ Ⓑ Ⓒ Ⓓ
22	One watt is equal to:	1 joule	1 newton metre	1 joule per second	1 newton per second	Ⓐ Ⓑ Ⓒ Ⓓ
23	A perpetual motion machine is impossible because it would violate the principle of:	conservation of energy	conservation of momentum	moments	inertia	Ⓐ Ⓑ Ⓒ Ⓓ
24	The idea that a scientific claim must be capable of being shown to be wrong by evidence is called:	verification	classification	observation	falsifiability	Ⓐ Ⓑ Ⓒ Ⓓ

SECTION B — Short Questions (3 × 3 = 9 Marks)

Q.2	Attempt all three parts.	Marks
(i)	The main scale of a vernier caliper has a smallest division of 1 mm and its vernier scale has 50 divisions. (a) Calculate the least count of the caliper in cm. [1] (b) While measuring a rod, the main scale reading is 6.7 cm and the 25th division of the vernier scale coincides with a main scale division. Find the length of the rod. [2]	**3**
(ii)	(a) Differentiate between the centre of mass and the centre of gravity of a body. [2] (b) State the condition under which these two points are *not* the same. [1]	**3**
(iii)	At sea level, atmospheric pressure supports a mercury column of height 76 cm in a barometer. The density of mercury is 13,600 kg/m^3. (a) If the same barometer is filled instead with a liquid of density 3,400 kg/m^3, calculate the height of the liquid column that the same atmospheric pressure would support. [2] (b) State how the height of the column depends on the density of the liquid used. [1]	**3**

SECTION C — Long Questions (2 × 5 = 10 Marks)

Q.3	Attempt both questions.	Marks
(i)	The speed-time graph below shows the motion of a body over a period of 40 seconds.	**5**

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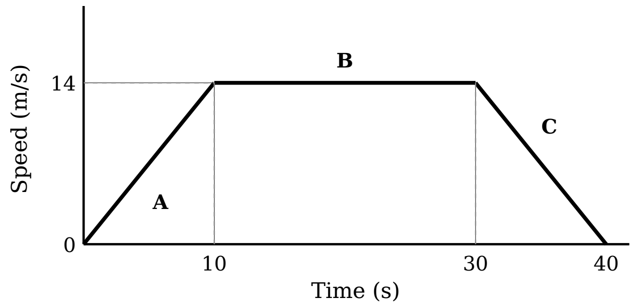

Figure 1: Speed-time graph of the body (stages A, B and C)
(a) Describe the motion of the body in each of the three stages A, B and C. [1]
(b) Calculate the acceleration of the body during stage A. [2]
(c) Using the area under the graph, calculate the total distance travelled by the body in the 40 seconds. [2]
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<td>(ii)</td>
<td>(a) Differentiate between renewable and non-renewable energy resources, naming two examples of each. [2]
(b) State one advantage and one disadvantage of generating electricity from a hydroelectric power station. [1]
(c) A conveyor belt moves a crate horizontally at a steady speed of 2 m/s against a total frictional force of 350 N. Calculate (i) the work done by the belt in 90 seconds, and (ii) the power output of the belt. [2]</td>
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Student's Signature: ____________________ Invigilator: ____________________ Date: ______________

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